

AREA II HYDROMETEOROLOGY/ HYDROLOGY

1. A stream that flows into a larger stream or other body of water.
 - a) Headwater
 - b) Tributary
 - c) Divide
 - d) Outlet
 - e) Water table wells
 - f) Shallow well
 - g) None of the above
2. Streamflow is:
 - a) Direct runoff + Baseflow
 - b) Direct runoff + Surface runoff
 - c) Baseflow + Interflow
 - d) Baseflow + Ground water flow
3. It show how rainfall is transformed into runoff
 - a) Hyetograph + Hydrograph
 - b) Hyetograph + Double mass curve
 - c) Hydrograph only
 - d) Hyetograph only
4. PAGASA color-coded rainfall classification measuring 15-30 mm.
 - a) Red
 - b) Orange
 - c) Yellow
 - d) Green
5. Which method is used to estimate missing rainfall data when the rainfall at the station of interest varies significantly from neighboring stations, by weighting the records using their long-term averages?
 - a) Arithmetic mean
 - b) Thiessen polygon
 - c) Isohyetal method
 - d) Normal ratio method
6. The probability that the event will occur at least once in n years.
 - a) Return period
 - b) Exceedance probability
 - c) Risk
 - d) Payback period
7. Baseflow is also called as:
 - a) Groundwater flow
 - b) Interflow
 - c) Overland flow
 - d) Superfluous water
 - e) None of the above
8. Well drilled into an artesian aquifer and water rises above the ground like a spring.
 - a) Free flowing well
 - b) Non-free flowing well
9. It shows value of stage versus discharge:
 - a) Hyetograph
 - b) Hydrograph
 - c) Stage
 - d) Rating curve
10. Time for water from farthest point in watershed to reach outlet.
 - a) Time to peak
 - b) Time lag
 - c) Time of concentration
 - d) Time base
11. A small watershed has a longest watercourse length of 1200 m and an average slope of 0.02 m/m. Using the Kirpich formula, estimate the time of concentration (T_c) in minutes.

$T_c = 0.0195 L^{0.77} = 0.75$

 - a) 20 minutes
 - b) 21 minutes
 - c) 22 minutes
 - d) 23 minutes
12. Moisture in this zone is less than saturation except during rainfall occurrence or irrigation.
 - a) Soil-water zone
 - b) Immediate vadose zone
 - c) Capillary zone
 - d) Hygroscopic zone
13. Portion of rainfall that produces direct runoff after losses such as infiltration, evaporation, interception, depression storage.
 - a) Effective rainfall
 - b) Interception
 - c) Throughfall
 - d) Through flow
14. The sinking of surface water into deeper layers, driven by wind convergence or higher water density at the surface. It influences heat distribution, circulation, and energy exchange in oceans, which are critical to global climate systems.
 - a) Surface tension
 - b) Capillary rise
 - c) Upwelling
 - d) Downwelling
 - e) None of the above

15. Rain falls through canopy to soil surface.

- a) Interflow
- b) Throughfall
- c) Interception
- d) Throughflow

16. The zone of saturation where all layers are filled with water under hydrostatic pressure.

- a) Vadose zone
- b) Phreatic zone
- c) Surface zone
- d) Water table

17. Gray or blue-gray layers that may cover the whole sky; sun appears dim or hidden. Precipitation likely to follow, especially if thickening.

- a) Altostratus (As)
- b) Altostratus (As)
- c) Cirrostratus (Cs)
- d) Cirrocumulus (Cc)

18. Soil water type that drains quickly and is unavailable to plants.

- a) Capillary water
- b) Gravitational water
- c) Hygroscopic water
- d) RAM

19. It is the volume of water that an aquifer releases from or takes into storage per unit surface area of aquifer per unit change in the component of head normal to that surface.

- a) Porosity
- b) Void ratio
- c) Specific retention
- d) Specific yield
- e) Storage coefficient

20. Layer of the atmosphere where all weather occurs:

- a) Troposphere
- b) Stratosphere
- c) Mesosphere
- d) Thermosphere
- e) All of the above

21. Earth emits solar radiation amount of:

- a) $\sim 390 \text{ W/m}^2$ upward
- b) $\sim 390 \text{ W/m}^2$ downward
- c) $\sim 370 \text{ W/m}^2$ upward
- d) $\sim 370 \text{ W/m}^2$ downward

22. Which type of evapotranspiration represents the maximum possible water loss from a surface, assuming unlimited water availability

- a) Reference ET
- b) Potential ET
- c) Actual ET
- d) Maximum ET

23. Inherent property of soil or rock to transmit water; it depends on pore size or connectivity.

- a) Infiltrability
- b) Permeability
- c) Sorptivity
- d) Hydraulic conductivity

24. Irregular, chaotic motion of air, usually caused by obstacles, surface heating, or wind shear, leading to eddies and gusts.

- a) Front
- b) Storm
- c) Turbulence
- d) Wind
- e) None of the above

25. Lead time for a TCWS # 3 is:

- a) 36hrs
- b) 24hrs
- c) 18hrs
- d) 16hrs
- e) 12hrs

26. Type of cloud that is white or gray patches in rounded masses or waves. Possible thunderstorms later in the day if humid.

- a) Altostratus (As)
- b) Altostratus (As)
- c) Cirrostratus (Cs)
- d) Cirrocumulus (Cc)

27-29. A 2-hectare rice field received 15 mm of rainfall in 30 minutes.

27. What is the rainfall intensity in mm/hr?

- a) 3.0 mm/hr
- b) 30.0 mm/hr
- c) 0.03 mm/hr
- d) 0.30 mm/hr

$$\frac{15 \text{ mm}}{30 \text{ min}} \times \frac{60 \text{ min}}{1 \text{ hr}} = 30 \text{ mm/hr}$$

28. How much water (in cubic meters) was added to the field?

- a) 3 m^3
- b) 30 m^3
- c) 300 m^3
- d) 0.3 m^3

$$20,000 \text{ m}^2 \times 0.015 \text{ m} = 300 \text{ m}^3$$

29. How many liters of water is this equivalent to?

- a) 300 liters
- b) 3000 liters
- c) 30000 liters
- d) 300000 liters

30. Solar constant:

- a) 1,637 W/m²
- b) 1,673 W/m²
- c) 1,376 W/m²
- d) 1,367 W/m²

31. Determine the volume of water release by lowering the piezometric surface of a confined aquifer by 5 meters over an area of $A = 1 \text{ km}^2$. Aquifer is 35-meter thick and has storage coefficient of 8.3×10^{-3} .

- a) 51 000 cu.m
- b) 45 000 cu.m
- c) 14 000 cu.m
- d) 41 500 cu.m
- e) None of the above

32. Rainfall intensity of 20mm/hr occurred over a watershed with an area of 100ha. The rainfall last for 6hrs. The direct runoff volume in the stream was measured and is equal to 30,000m³. The precipitation not available to runoff in this case is:

- a) 9cm
- b) 3cm
- c) 17.5mm
- d) 5mm
- e) None of the above

33. A 20 cm well penetrates 30 m below static water level (ground water table level). After a long period of pumping at a rate of 1800 lpm, if the drawdowns in the observation wells at 12 m and 36 m from the pumped well are 1.2 m and 0.5 m, respectively. Calculate the transmissibility of the aquifer.

- a) 666.6 m²/day
- b) 666.3 m²/day
- c) 663.6 m²/day
- d) 663.3 m²/day

34. A 20-ft thick confined aquifer with hydraulic conductivity of 400 ft/day was tapped by a 4-in diameter STW. With a radius of influence of 2500 ft, determine the maximum discharge of the STW in lps. Assume an allowable drawdown of 12 ft.

- a) 20.36 lps
- b) 20.46 lps
- c) 20.66 lps
- d) 20.56 lps

35. The portion of soil water that plants can use without experiencing stress is approximately equal to:

- a) 75% of AWC
- b) 50% of AWC
- c) 70% AWC

d) 55% AWC

36. Measures the water bearing capacity of a geologic formation.

- a) Porosity
- b) Void ratio
- c) Specific retention
- d) Specific yield
- e) Storage coefficient

37. A line that connects points of equal elevation above a reference level, most often sea level.

- a) Isohyet
- b) Isobar
- c) Contour
- d) Streamline
- e) None of the above

38. Are a unit used to describe the fraction of the sky covered by clouds, observed from a specific location.

- a) Oktas
- b) Oskat
- c) Ostak
- d) Otkas

39. Shortest day in Northern Hemisphere; Sun overhead at Tropic of Capricorn (23.5°S).

- a) Equinox
- b) Summer solstice
- c) Winter solstice
- d) Eclipse

40. A real body that absorbs/ emits a constant fraction of radiation across all wavelengths.

- a) Blackbody
- b) Graybody
- c) Emitter
- d) Absorber

41. These are gauges that measures the water level from above the surface such as bridges or other structure.

- a) Staff gauges
- b) Wire-weight gauges
- c) Float-gauge recorder
- d) Stage recorder

42. A lake had a water surface elevation of 103.2 m above datum at the beginning of the month. At the same month, lake received an average inflow of 6 cms. from surface runoff sources. For the same period, the outflow from the lake had an average value of 6.5 cms. Further, in the same month the lake received a rainfall of 145 mm and losses due to evaporation of 6.1 cm. Assume the area of the

$$Q = \frac{2\pi K b (h_2 - h_1)}{\ln\left(\frac{r_2}{r_1}\right)}$$
$$= \frac{2\pi (400) (20) (12)}{\ln\left(\frac{2500}{3.28}\right)} = 20.55$$

$$\left(\frac{145}{24 \times 60}\right) \left(\frac{145}{24 \times 60}\right) \left(\frac{1}{3.28}\right)^3 \left(\frac{1000}{1 \text{ m}^3}\right) =$$

$$MAP = 10$$
$$PM = 8$$

lake is 5000 has. and assume no contribution from ground water storage. What is the water surface elevation of the lake after the month?

- a) 103.301m
- b) 103.401m
- ☒ c) 103.258m
- d) 102.258m

43. Two observation wells penetrating into a confined aquifer are located 1.5 km apart in the direction of flow. Heads of 45 m and 20 m are indicated at these two observation wells. If the coefficient of permeability of the aquifer is 30 m/day and the porosity is 0.25, determine the time of travel of an inert tracer from one well to another.

- a) 745 days
- ☒ b) 750 days
- c) 755 days
- d) 735 days

44. A 16-ft thick confined aquifer with hydraulic conductivity of 500 ft/day was tapped by a 4 in diameter shallow tube well. With a radius of influence of 2000 ft. Determine the maximum discharge of STW in lps. Assume an allowable drawdown of 10 ft.

- a) 17.45 lps
- b) 17.44 lps
- ☒ c) 17.54 lps
- d) 17.65 lps

45. Wind threat for TCWS No. 3:

- a) 39-61 kph
- b) 62-88 kph
- ☒ c) 89-117 kph
- d) 118-184 kph

46. TCWS No. 4 cyclone category:

- a) Tropical depression
- b) Tropical storm
- ☒ c) Typhoon
- d) Severe tropical storm

47. In calculating ET using FAO Penman-Monteith model, wind speed measurement should be taken _____ above the ground.

- a) 10m
- b) 1m
- ☒ c) 2m
- d) 5m
- e) None of the above

48. PAGASA color-coded rainfall classification measuring 4 to 8 gallons per square meter/hour.

- a) Red
- ☒ b) Orange

- c) Yellow
- d) Black

49. When droplets or crystals become heavy enough to overcome updrafts, they fall as:

- a) Runoff
- b) Infiltration
- ☒ c) Precipitation
- d) Virga

50. Philippine Climate Zones: No pronounced dry season; heavy rainfall typically from November to January. Characteristics of eastern Philippines (e.g. Eastern Samar).

- a) Type I
- ☒ b) Type II
- c) Type III
- d) Type IV

51. Find the 4th term of $(2x + 3y)^5$.

A. $2x^3y^2$

B. $10x^3y^2$

☒ C. $720x^2y^3$

D. None of the above

52. In a series 3, 6, 12, 21 ... what is the 15th term?

A. 87

B. 168

☒ C. 318

D. 411

53. Miriam can finish a job in 5 hours, Jorge can finish the same job in 3 hours. How long will it take them to finish the whole job if they work together?

A. $3/4$ hour

B. $5/4$ hours

☒ C. $15/8$ hours

D. $8/3$ hours

54. It takes 6 hours for pump A, used alone, to fill a tank of water. Pump B used alone takes 8 hours to fill the same tank. We want to use three pumps: A, B and another pump C to fill the tank in 2 hours. How long would

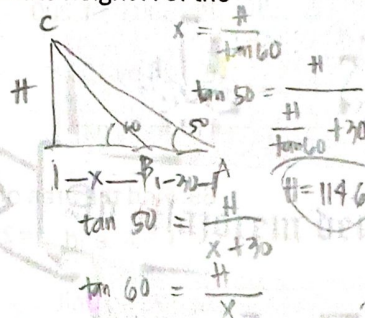
it take pump C, used alone, to fill the tank?

- a. 5 hours
- b. 8.5 hours
- c. 4 hours & 30 minutes
- d. 4 hours & 48 minutes

1.8
4 hrs 48 mins

55. The angle of elevation to the top C of a building from two points A and B on level ground are 50 degrees and 60 degrees respectively. The distance between points A and B is 30 meters. Find the approximate height H of the building.

- A. 95 meters
- B. 100 meters
- C. 110 meters
- D. 115 meters



56. Find the equation of the line passing through (4, 0) and is parallel to the line $x - y - 2 = 0$.

- A. $y + x + 4 = 0$
- B. $y - x + 4 = 0$
- C. $y - x - 4 = 0$
- D. $y + x - 4 = 0$

$y = x - 2$
 $m = 1$
 $b = -2$
 $y - y_1 = m(x - x_1)$
 $y - 0 = 1(x - 4)$
 $y = x - 4$
 $y - x + 4 = 0$

57. Find the equation of the line passing through (6, -3) and (2, 0).

- A. $4x - 3y = 6$
- B. $4x + 3y = 6$
- C. $3x - 4y = 6$
- D. $3x + 4y = 6$

$y = mx + b$
 $6, -3 \rightarrow 0 = \frac{3}{4}(2) + b$
 $2, 0 \rightarrow b = -1.5$
 $m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{0 - (-3)}{2 - 6} = -\frac{3}{4}$
 $y - 0 = -\frac{3}{4}(x - 2)$

58. Determine the radius of the circle whose equation

$x^2 + y^2 - 12x + 20y + 15 = 0$

- A. 121
- B. 100
- C. 10
- D. 11

$(x^2 - 12x) + (y^2 + 20y) = -15$
 $4(x - 3)^2 + (y + 10)^2 = 121$
 $\sqrt{121} = 11$

59. The length of the major axis of the ellipse with equation $4x^2 + 9y^2 = 36$ is

- A. 3
- B. 2
- C. 4
- D. 6

$4x^2 + 9y^2 = 36$
 $\frac{x^2}{9} + \frac{y^2}{4} = 1$
6

60. The vertex of the parabola $x^2 - 4x - 8y - 28 = 0$ is at:

- A. (2, 4)
- B. (2, -4)
- C. (4, 2)
- D. (-4, 2)

$x^2 - 4x - 8y - 28 = 0$
 $x^2 - 4x - 8y = 28$
 $x^2 - 4x + 4 = 8y + 32$
 $(x - 2)^2 = 8(y + 4)$
Vertex: (2, -4)

61. Find the center of the hyperbola $16y^2 - 9x^2 - 64y - 80 = 0$.

- A. (2, 0)
- B. (0, 2)
- C. (0, -2)
- D. (-2, 0)

$16y^2 - 9x^2 - 64y - 80 = 0$
 $16(y^2 - 4y) - 9x^2 - 80 = 0$
 $16(y^2 - 4y + 4) - 9x^2 - 80 + 64 = 0$
 $16(y - 2)^2 - 9x^2 - 16 = 0$
 $\frac{(y - 2)^2}{1} - \frac{x^2}{\frac{16}{9}} = 1$
Center: (0, 2)

62. A ladder 10 feet long rests against a vertical wall. If the bottom of the ladder slides away from the wall at a rate 1 ft/sec, how fast is the top of the ladder sliding down the wall when the bottom of the ladder is 6 feet from the wall?

- A. 0.75 ft/sec
- B. -0.75 ft/sec
- C. 0.5 ft/sec
- D. -0.5 ft/sec

63. Water flows into a vertical cylindrical tank at 12 ft³/min, the surface rises 6 in/min. Find the radius of the tank.

- A. 2.8 ft
- B. 1.8 ft
- C. 3.8 ft
- D. 4.8 ft

$\frac{dV}{dt} = 12 \text{ ft}^3/\text{min}$
 $\frac{dh}{dt} = 6 \text{ in/min} = 0.5 \text{ ft/min}$
 $V = \pi r^2 h$
 $\frac{dV}{dt} = \pi r^2 \frac{dh}{dt}$
 $12 = \pi r^2 (0.5)$
 $r^2 = \frac{12}{0.5\pi} = \frac{24}{\pi}$
 $r = \sqrt{\frac{24}{\pi}} \approx 2.77 \text{ ft}$

64. A water tank has the shape of an inverted circular cone with the base radius 2 meters and height 4 meters. If

water is being pumped into the tank at a rate of 2 m/minute, find the rate at which the water level is rising when the water is 3 meters deep.

- A. -0.28/min
- ☒ B. 0.28 m/min
- C. 0.4m/min
- D. -0.4 m/min

65. The cost C of a product is a function of the quantity x of the product is given by the relation: $C(x) = x^2 - 4000x + 50$. Find the quantity for which the cost is minimum.

- A. 3000
- ☒ B. 2000
- C. 1000
- D. 1500

$$C(x) = x^2 - 4000x + 50$$

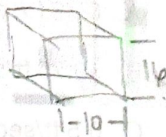
$$2x - 4000 = 0$$

$$\frac{2x}{2} = \frac{4000}{2}$$

$$x = 2000$$

66. By cutting away identical squares from each corner of a rectangular cardboard, and folding up the resulting flaps, the cardboard may be turned into an open box. If the cardboard is 16 inches long and 10 inches wide, determine the dimensions of the box that will yield the maximum volume. What is the greatest volume?

- A. 140 cubic inches
- ☒ B. 144 cubic inches
- C. 148 cubic inches
- D. 150 cubic inches



67. Under a perfect competition, a firm can sell at a price of P200 per unit in all of a particular commodity it produces. If $C(x)$ pesos is the total cost of each day's production when x units are produced, and $C(x) = 2x^2 + 40x + 1400$, find the number of units that should be produced daily for the firm to have the greatest daily total profit.

- A. 30 units
- B. 35 units
- ☒ C. 40 units
- D. 45 units

$$200x$$

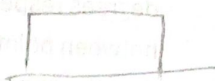
$$C(x) = 2x^2 + 40x + 1400$$

$$4x + 40$$

$$\frac{4x}{4} = \frac{-40}{4} \quad x(-10)$$

68. A farmer has 2,400 feet of fencing and wants to fence off a rectangular field that borders a straight river. He needs no fence along the river. What are the dimensions of the field that has the largest area?

- A. 60 by 120
- ☒ B. 600 by 1,200
- C. 24 by 100
- D. 10 by 240



69. Find the area of the region bounded by $y = x^2$ and $y = 2x - x^2$

- A. $2/3$ sq unit
- B. $3/8$ sq unit
- ☒ C. $1/3$ sq unit
- D. $5/8$ sq unit

$$y = x^2$$

$$y = 2x - x^2$$

$$x^2 = 2x - x^2$$

$$0 = 2x - x^2 - x^2$$

$$= 2x - 2x^2$$

$$\frac{2x^2}{2} = \frac{-2x}{2}$$

70. Two months ago there were 5 mice, now there are 18. Assuming the growth continues like that, how many mice will there be two months from now?

- A. 44
- ☒ B. 65
- C. 98
- D. 123

$$m(t) = pe^{kt}$$

$$5e$$

71. After 500 years, a sample of radium-226 has decayed to 80.4% of its original mass. Find the half-life of radium-226.

- A. 1590 years
- ☒ B. 1595 years
- C. 1598 years
- D. 2000 year

$$80.4 =$$

72. The mass of a radium 226 is given by $m(t) = \beta e^{-kt}$. Its half-life is 1,590 years. A sample of a radium-226 has a mass of 1,000 mg. Find the mass after 1,000 years.

- A. 650 mg
- B. 100 mg
- C. 165 mg
- D. 200 mg

73. A 10-meter tape is 5 mm short. What is the correct length in meters?

- A. 9.995 m
- B. 10.05 m
- C. 9.95 m
- D. 10.005 m

74. The distance between two points measured with a steel tape was recorded as 916.58 ft. Later, the tape was checked and found out to be only 99.9 ft long. What is the true distance between the points?

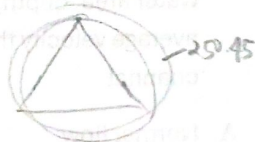
- A. 935.66 ft
- B. 966.15 ft
- C. 955.66 ft
- D. 915.66 ft

75. 1° of Earth's rotation is equivalent to _____.

- A. 1 hour
- B. 4 minutes
- C. 30 minutes
- D. 360 seconds

76. The area of a circle circumscribing about an equilateral triangle is 250.45 square meters. What is the area of the triangle?

- A. 93.6 m²
- B. 103.6 m²
- C. 113.6 m²
- D. 123.6 m²



77. The pressure under the gravity dam producing an overturning effect.

- A. Uplift pressure
- B. Downward pressure
- C. Foundation pressure
- D. None of the above

78. Type of flow wherein the depth of flow in a prismatic channel is constant at any point in the channel.

- A. laminar flow
- B. uniform flow
- C. steady flow
- D. critical flow

79. Type of flow wherein the discharge or flow in a prismatic channel is constant with respect to time.

- a. laminar flow
- b. uniform flow
- C. steady flow
- D. critical flow

80. A turbine is rated at 600 hp when the flow of water through it is 0.61 m³/s. Assuming an efficiency of 87%, what is the head acting on the turbine?

- a. 82.3 m
- b. 86 m
- C. 84.1 m
- D. 90.5 m

81. It is derived from the principle of conservation of energy.

- a. Darcy's Equation
- b. Manning's Equation
- c. Bernoulli's Equation
- D. Hagen-Poiseuille

82. The theoretical flow velocity in an orifice is 4 m/s, what is the height of water flowing above the center of the orifice?

- a. 0.8 m
- b. 8.0 m
- C. 0.9 m
- D. 9.0 m

$$V = \sqrt{2gh}$$

83. Calculate the discharge through 100-mm diameter orifice under a head of 5.5 m of water.

Assume $C_c = 0.61$ and $C_v = 0.98$.

- A. $4.87 \text{ m}^3/\text{hr}$ C. $48.7 \text{ m}^3/\text{hr}$
B. 4.87 lps D. 48.7 lps

84. For a 2-inch orifice with a discharge coefficient of 0.60, what head results when measuring a discharge of 0.45 cfs?

- a. 6.50 m c. 5.60 m
b. 4.60 m d. 4.30 m

85. A barrier placed on a stream/canal to constrict the flow of water and cause it fall over a crest, usually used to measure discharge.

- a. drop structure
b. Weir
c. crest stage gage
D. staff gage

86. A concrete culvert 1.2 m in diameter and 5 m long conveys flood water. Both ends of the culvert are submerged and the difference in water level upstream and downstream is 2.40 m. Calculate the discharge assuming $C = 0.61$.

- A. $1.235 \text{ m}^3/\text{s}$ C. $2.682 \text{ m}^3/\text{s}$
B. $4.734 \text{ m}^3/\text{s}$ D. $0.876 \text{ m}^3/\text{s}$

87. The Reynold's number is a relevant number when describing:

- a. Discharge through open channels
b. Friction over the pipe
c. Depth of Flow
d. Characteristic of the flow

88. The Reynold's number for a laminar flow is

- a. less than 2000
b. greater than 2000

C. greater than 800

D. less than 800

89. Water is flowing in a pipe with a radius of 25.4 cm at a velocity of 5 m/s at the temperature in the pipe. The density and viscosity of the water are as follows: density = 997.9 kg/m^3 and viscosity = $1.131 \text{ Pa}\cdot\text{s}$. What is the Reynolds number for this situation?

- a. 2241 C. 3100
b. 96.2 D. 1140

90. Determine the friction factor of a laminar pipe flow if the Reynolds number is 1700.

- a. 0.05 C. 0.04
b. 0.08 D. 0.20

91. When the water level in the pipe drops below the upper edge of the pipe or culvert and its conduit controls, the flow becomes:

- A. weir flow C. orifice flow
B. open channel flow D. pipe flow

92. A conduit in which a liquid flows with a free surface.

- A. Open Channel C. Canal
B. Pipe Flow D. Waterways

93. Slope of the water surface profile plus the velocity head in open channels.

- A. Energy grade line slope
B. Water surface slope
C. Channel bottom slope
D. Hydraulic grade line slope

94. Occurs when flow has a constant water area, depth, discharge, and average velocity through a reach of channel

- A. Normal flow C. Uniform Flow
B. Critical flow D. Varied flow

95. Constant flow depth along a longitudinal section of a channel under a uniform flow condition

- A. Normal depth C. Uniform depth
B. Critical depth D. None of the above

96. Water flows uniformly in a rectangular, concrete, open channel that is 10 m wide at a depth of 3 m. The channel slope is 0.0025. Using $n=0.013$, find the velocity.

- A. 9.848 m/s
B. 7.848 m/s
C. 5.848 m/s
D. 3.848 m/s

$$V = \frac{1}{0.013} \left[\frac{(10)(3)}{10+2(3)} \right]^{2/3} (0.0025)^{1/2}$$

5.848

97. Determine the flow rate of the previous item.

- A. 135.44 m³/s
B. 155.44 m³/s
C. 175.44 m³/s
D. 185.44 m³/s

$$Q = VA$$

98. Determine the boundary shear stress of the previous item.

- A. 46 Pa
B. 48 Pa
C. 50 Pa
D. 52 Pa

99. A rectangular, concrete channel, 15 m wide is to carry water at the rate 22 m³/s. If the channel slope is 0.00025, determine the normal depth of flow.

Use roughness coefficient $n=0.013$.

- A. 1.1 m
B. 1.2 m
C. 1.3 m
D. 1.4 m

$$22 = \left[\frac{1}{0.013} \left(\frac{15 \times d}{15+2d} \right)^{2/3} (0.00025)^{1/2} \right]$$

[15 x d]
1.18
1.2

100. Determine the uniform flow (m³/s) through a trapezoidal concrete lined canal having side slope of 3H to 4V and bottom width of 2 m if the depth of flow is 2 m. The channel is laid on a slope of 3 m per 2 kilometers. Use $n=0.013$.

- A. 20.85 m³/s
B. 30.85 m³/s
C. 40.85 m³/s
D. 50.85 m³/s

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